2.50 The Normal Distribution

Worked Solutions

1)

A is $X \sim \mathrm{N}(10, 3)$

B is $X \sim \mathrm{N}(20, 6)$

C is $X \sim \mathrm{N}(20, 3)$

D is $X \sim \mathrm{N}(10, 1)$

2)

Mean is 13

Standard Deviation is 2.5

3)

Mean is 44

Standard Deviation is 5

4 a)

$\sigma = \sqrt{49} = 7$

4 b)

$\sigma = \sqrt{5}$

4 c)

$\sigma = \sqrt{\sqrt{2}} = \sqrt[4]{2}$

5 a)

$\mu = 30,\ \sigma = \sqrt{25} = 5$

$a = 30 - 3 \times 5 = 15$

$b = 30 + 3 \times 5 = 45$

5 b)

$\mu = 18,\ \sigma = \sqrt{12.25} = 3.5$

$a = 18 - 3 \times 3.5 = 7.5$

$b = 18 + 3 \times 3.5 = 28.5$

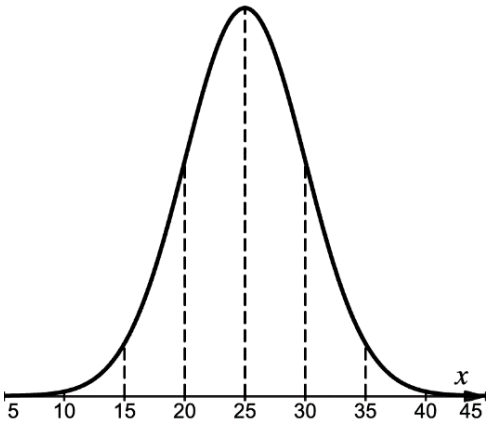
5 c)

$\mu = 605,\ \sigma = \sqrt{1024} = 32$

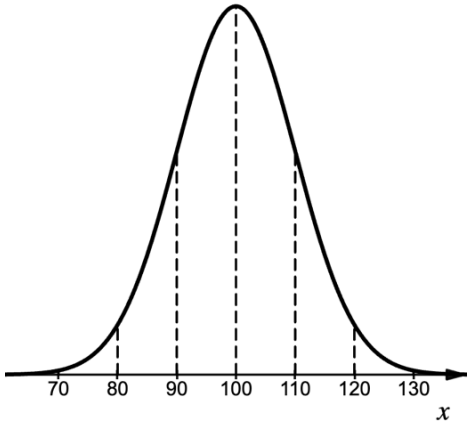
$a = 605 - 3 \times 32 = 509$

$b = 605 + 3 \times 32 = 701$

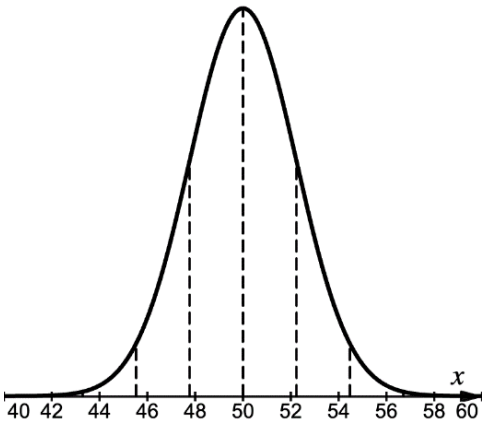
6 a)


x
5 10 15 20 25 30 35 40 45

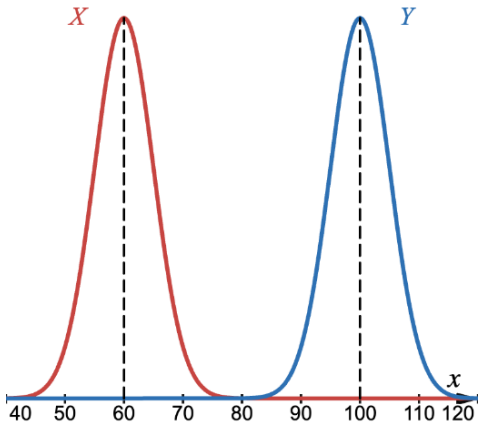
6 b)


70 80 90 100 110 120 130
x

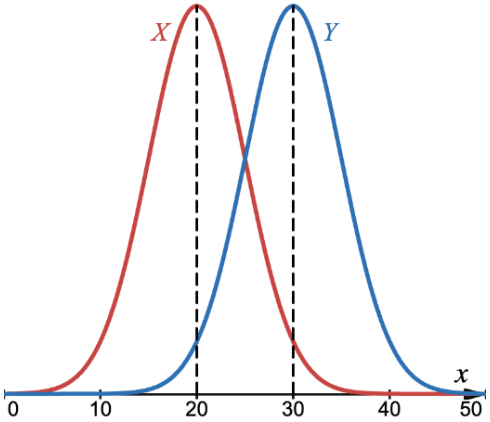
6 c)


x
40 42 44 46 48 50 52 54 56 58 60

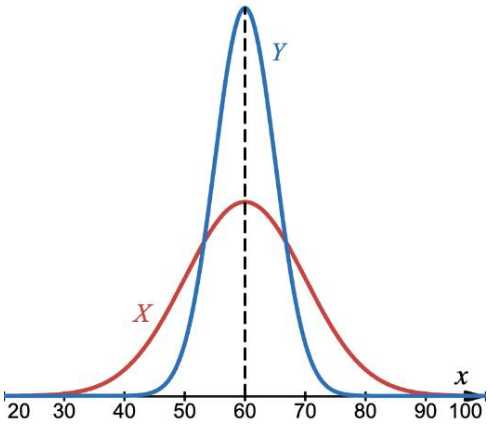
7)


X
Y
x
40 50 60 70 80 90 100 110 120

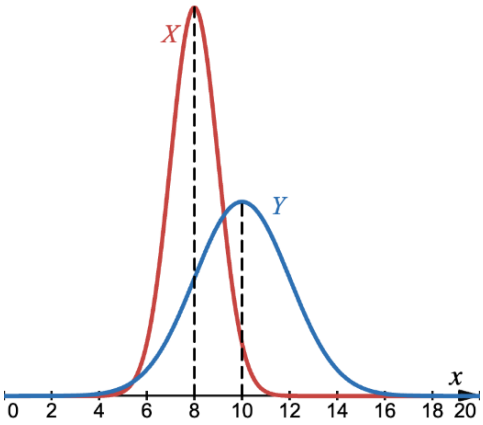
8)


X
Y
x
0 10 20 30 40 50

9)


Y
X
x
20 30 40 50 60 70 80 90 100

10)



11 a)

$\mathrm{P}(X < 68) = 1 - 0.1587 = 0.8413$

11 b)

$\mathrm{P}(X < 60) = 0.1587$

11 c)

$\mathrm{P}(60 < X < 68) = 1 - 2(0.1587) = 0.6826$

12 a)

$\mathrm{P}(X < 80) = \frac{1-0.682690}{2} = 0.1587$

12 b)

$\mathrm{P}(X > 120) = 0.1587$

12 c)

$\mathrm{P}(80 < X < 100) = \frac{0.682690}{2} = 0.3413$

13 a)

Using $Z = \frac{X-\mu}{\sigma}$:

$Z = \frac{7-8}{2} = -0.5$

13 b)

$Z = \frac{9-8}{2} = 0.5$

13 c)

$Z = \frac{10.2-8}{2} = 1.1$

14 a)

$Z = \frac{15-16}{4} = -0.25$

$\mathrm{P}(X < 15) = \mathrm{P}(Z < -0.25) = 0.4013$

14 b)

$Z = \frac{17-16}{4} = 0.25$

$\mathrm{P}(X > 17) = \mathrm{P}(Z > 0.25) = 0.4013$

14 c)

$Z = \frac{13-16}{4} = -0.75$

$Z = \frac{15-16}{4} = -0.25$

$\mathrm{P}(13 < X < 15) = \mathrm{P}(-0.75 < Z < -0.25) = 0.1747$

14 d)

$Z = \frac{18-16}{4} = 0.5$

$Z = \frac{19-16}{4} = 0.75$

$\mathrm{P}(18 < X < 19) = \mathrm{P}(0.5 < Z < 0.75) = 0.0819$

15 a)

$\mathrm{P}(X \leq 37) = 0.3538$

15 b)

$\mathrm{P}(X < 37) = 0.3538$

15 c)

$\mathrm{P}(X \geq 45) = 0.2660$

15 d)

$\mathrm{P}(X > 45) = 0.2660$

16 a)

$\mathrm{P}(X < 860) = 0.9088$

16 b)

$\mathrm{P}(X > 810) = 0.4121$

16 c)

$\mathrm{P}(780 < X < 850) = 0.5384$

16 d)

$\mathrm{P}(860 < X < 880) = 0.0535$

17 a)

$\mathrm{P}(X = 46) = 0$

17 b)

$\mathrm{P}(X \neq 40) = 1$

18 a)

$\mathrm{P}(X < 80) = 0.5$

18 b)

$\mathrm{P}(X < 75) = 0.1807$

18 c)

$\mathrm{P}(X > 83) = 0.2919$

18 d)

$\mathrm{P}(75 < X < 83) = 0.5274$

19 a)

$\mathrm{P}(X < 110) = 0.0549$

19 b)

$\mathrm{P}(125 < X < 130) = 0.2030$

19 c)

$\mathrm{P}(130 < X < 135) = 0.1538$

20 a)

$\mathrm{P}(-1 < Z < 1) = 0.6827$

20 b)

$\mathrm{P}(-2 < Z < 2) = 0.9545$

20 c)

$\mathrm{P}(-3 < Z < 3) = 0.9973$

20 d)

$\mathrm{P}(-4 < Z < 4) = 0.9999$

21 a)

$\mathrm{P}(185 < X < 215) = 0.6827$

21 b)

$\mathrm{P}(170 < X < 230) = 0.9545$

21 c)

$\mathrm{P}(155 < X < 245) = 0.9973$

21 d)

$\mathrm{P}(140 < X < 260) = 0.9999$

22)

$Z = \frac{0.9\mu - \mu}{\frac{1}{2}\mu} = -0.2$

$\mathrm{P}(X < 0.9\mu) = \mathrm{P}(Z < -0.2) = 0.4207$

23)

$Z = \frac{\frac{11}{2}\mu - 5\mu}{4\mu} = \frac{1}{8}$

$\mathrm{P}(X > \frac{11}{2}\mu) = \mathrm{P}(Z > \frac{1}{8}) = 0.4503$

24)

$Z = \frac{0.1\mu - 2\mu}{9\mu} = -\frac{19}{90}$

$Z = \frac{0.5\mu - 2\mu}{9\mu} = -\frac{1}{6}$

$\mathrm{P}(0.1\mu < X < 0.5\mu) = \mathrm{P}(-\frac{19}{90} < Z < -\frac{1}{6}) = 0.0174$

25 a)

$\bar{x} = \frac{\sum x}{200} = \frac{105000}{200} = 525$

$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{440000}{199} = 2211.055276$

$s = \sqrt{\frac{440000}{199}} = 47.0$

25 b)

$X \sim \mathrm{N}(525, 2211.055276)$

$\mathrm{P}(X > 535) = 0.4158$

26 a)

$\bar{x} = \frac{\sum x}{100} = \frac{4650}{100} = 46.5$

26 b)

$s^2 = \frac{\sum x^2}{n-1} - \frac{(\sum x)^2}{n(n-1)} = \frac{219000}{99} - \frac{4650^2}{100 \times 99} = \frac{925}{33} \approx 28.03$

26 c)

$X \sim \mathrm{N}(46.5, 28.03)$

$\mathrm{P}(43 < X < 46) = 0.2081$

27)

$2.01 - 3 \times 0.1 = 1.71$

$2.01 + 3 \times 0.1 = 2.31$

If a normal distribution was appropriate, we would expect 99% to be approximately between 1.71 m and 2.31 m. If we use $X \sim \mathrm{N}(2.01, 0.1^2)$ then $\mathrm{P}(1.83 < X < 2.19) \approx 93\%$ so it does not appear to be suitable.

28 a)

$a = -0.674$

28 b)

$b = 1.28$

28 c)

$c = 2.33$

28 d)

$d = -0.842$

29 a)

$a = 18.8$

29 b)

$b = 29.1$

29 c)

$c = 30.2$

29 d)

$d = 26.0$

30)

$\mathrm{P}(X < a) = 0.05 \Rightarrow a = 751$

$\mathrm{P}(X > b) = 0.05 \Rightarrow b = 849$

31)

$\mathrm{P}(X < 64) = 0.386415$

$\mathrm{P}(X < a) = 0.386415 + 0.25 \Rightarrow a = 66.2$

32 a)

$a = 19.7$

32 b)

$\mathrm{P}(X < c) = \frac{1-0.3}{2} = 0.35 \Rightarrow c = 16.78151$

$18 - b = 16.78151 \Rightarrow b = 1.21849 \approx 1.22$

33 a)

$\mu = 40$

33 b)

$\mathrm{P}(Z < b) = 0.3 \Rightarrow b = -0.5244005$

$-0.5244005 = \frac{50-\mu}{10} \Rightarrow \mu = 55.2$

33 c)

$\mathrm{P}(Z > c) = 0.7 \Rightarrow c = -0.5244005$

$-0.5244005 = \frac{200-\mu}{10} \Rightarrow \mu = 205$

33 d)

$\mathrm{P}(Z > d) = 0.01 \Rightarrow d = 2.326348$

$2.326348 = \frac{70-\mu}{10} \Rightarrow \mu = 46.7$

34 a)

$\mathrm{P}(Z < a) = 0.2 \Rightarrow a = -0.8416212$

$-0.8416212 = \frac{30-33}{\sigma} \Rightarrow \sigma = 3.56$

34 b)

$\mathrm{P}(Z < b) = 0.6 \Rightarrow b = 0.2533471$

$0.2533471 = \frac{50-33}{\sigma} \Rightarrow \sigma = 67.1$

34 c)

$\mathrm{P}(Z > c) = 0.99 \Rightarrow c = -2.326348$

$-2.326348 = \frac{1-33}{\sigma} \Rightarrow \sigma = 13.8$

34 d)

$\mathrm{P}(Z > d) = 0.49 \Rightarrow d = 0.02506891$

$0.02506891 = \frac{34-33}{\sigma} \Rightarrow \sigma = 39.9$

35)

$\mathrm{P}(Z < a) = 0.35 \Rightarrow a = -0.3853205$

$-0.3853205 = \frac{700-725}{\sigma} \Rightarrow \sigma = 64.9$

36)

$\mathrm{P}(Z < a) = \frac{1-0.5}{2} = 0.25 \Rightarrow a = -0.6744898$

$-0.6744898 = \frac{2.6-3}{\sigma} \Rightarrow \sigma = 0.593$ g

37)

$\mathrm{P}(Z < a) = 0.9 \Rightarrow a = 1.281552$

$1.281552 = \frac{40-18}{\sigma} \Rightarrow \sigma = 17.166685$

$18 - 3 \times 17.166685 = -33.500055$

$\therefore$ a large proportion of ambulance response times would be negative, so they are unlikely to be normally distributed.

38 a)

$\mu = 72.02721 + \frac{106.3728-72.02721}{2}$

$= 72.02721 + 17.172795 = 89.200005 \approx 89.2$

$\mathrm{P}(Z < a) = 0.1 \Rightarrow a = -1.281552$

$-1.281552 = \frac{72.02721-89.200005}{\sigma}$

$\Rightarrow \sigma = 13.3999986 \approx 13.4$

So $\sigma^2 = 179.5599624 \approx 179.56$

38 b)

$3 \times 89.200005 - 2 = 265.600015 \approx 265.6$

$3 \times 13.3999986 = 40.1999958 \approx 40.2$

$40.1999958^2 = 1616.039662 \approx 1616.04$

$\therefore Y \sim \mathrm{N}(265.6, 1616.04)$

39)

$326.8 = a \times 84 + b$

$\sqrt{72.4201} = a \times \sqrt{5.29} \Rightarrow a = 3.7$

$\therefore 326.8 = 3.7 \times 84 + b \Rightarrow b = 16$

40)

$\mathrm{P}(Z < a) = 0.5 \Rightarrow a = 0$

$0 = \frac{99-\mu}{\sigma} \Rightarrow \mu = 99$

$\mathrm{P}(Z > b) = 0.6 \Rightarrow b = -0.2533471$

$-0.2533471 = \frac{90-99}{\sigma} \Rightarrow \sigma = 35.5$

41)

$\mathrm{P}(Z > a) = 0.5 \Rightarrow a = 0$

$0 = \frac{11-\mu}{\sigma} \Rightarrow \mu = 11$

$\mathrm{P}(Z > b) = 0.05 \Rightarrow b = 1.644854$

$1.644854 = \frac{12-11}{\sigma} \Rightarrow \sigma = 0.608$ to 3sf

42 a)

$\mathrm{P}(Z < a) = 0.3 \Rightarrow a = -0.5244005$

$-0.5244005 = \frac{10-\mu}{\sigma}$

$\Rightarrow \mu - 0.5244005\sigma = 10 \quad (1)$

$\mathrm{P}(Z < b) = 0.35 \Rightarrow b = -0.3853205$

$-0.3853205 = \frac{11-\mu}{\sigma}$

$\Rightarrow \mu - 0.3853205\sigma = 11 \quad (2)$

$\therefore \mu = 13.8,\ \sigma = 7.19$

42 b)

$\mathrm{P}(Z < a) = 0.4 \Rightarrow a = -0.2533471$

$-0.2533471 = \frac{50-\mu}{\sigma}$

$\Rightarrow \mu - 0.2533471\sigma = 50 \quad (1)$

$\mathrm{P}(Z < b) = 0.65 \Rightarrow b = 0.3853205$

$0.3853205 = \frac{90-\mu}{\sigma}$

$\Rightarrow \mu + 0.3853205\sigma = 90 \quad (2)$

$\therefore \mu = 65.9,\ \sigma = 62.6$

42 c)

$\mathrm{P}(Z < a) = 0.01 \Rightarrow a = -2.326348$

$-2.326348 = \frac{3-\mu}{\sigma}$

$\Rightarrow \mu - 2.326348\sigma = 3 \quad (1)$

$\mathrm{P}(Z > b) = 0.02 \Rightarrow b = 2.053749$

$2.053749 = \frac{9-\mu}{\sigma}$

$\Rightarrow \mu + 2.053749\sigma = 9 \quad (2)$

$\therefore \mu = 6.19,\ \sigma = 1.37$

42 d)

$\mathrm{P}(Z > a) = 0.44 \Rightarrow a = 0.1509692$

$0.1509692 = \frac{67-\mu}{\sigma}$

$\Rightarrow \mu + 0.1509692\sigma = 67 \quad (1)$

$\mathrm{P}(Z > b) = 0.33 \Rightarrow b = 0.4399132$

$0.4399132 = \frac{73-\mu}{\sigma}$

$\Rightarrow \mu + 0.4399132\sigma = 73 \quad (2)$

$\therefore \mu = 63.9,\ \sigma = 20.8$

43)

$\mu = \frac{140+215}{2} = 177.5$

$\mathrm{P}(Z < a) = 0.14 \Rightarrow a = -1.080319$

$-1.080319 = \frac{140-177.5}{\sigma}$

$\Rightarrow \sigma = 34.71196933$

$X \sim \mathrm{N}(177.5, 34.71196933^2)$

$177.5 - 3 \times 34.71196933 \approx 73.4$ grams

44)

$y = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{2\pi}} \times -x \times \mathrm{e}^{-\frac{1}{2}x^2}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\sqrt{2\pi}} x \mathrm{e}^{-\frac{1}{2}x^2}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{\sqrt{2\pi}} x \times -x \times \mathrm{e}^{-\frac{1}{2}x^2} - \frac{1}{\sqrt{2\pi}} \times \mathrm{e}^{-\frac{1}{2}x^2}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{1}{\sqrt{2\pi}} x^2 \mathrm{e}^{-\frac{1}{2}x^2} - \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2}$

Points of Inflection exist when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$:

$\frac{1}{\sqrt{2\pi}} x^2 \mathrm{e}^{-\frac{1}{2}x^2} - \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2} = 0$

$\Rightarrow \frac{1}{\sqrt{2\pi}} x^2 - \frac{1}{\sqrt{2\pi}} = 0$

$\Rightarrow x^2 - 1 = 0 \Rightarrow x^2 = 1$

$\Rightarrow x = \pm 1$

45)

$y = \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sigma\sqrt{2\pi}} \times \left(-\frac{1}{2} \times 2 \times \frac{1}{\sigma}\left(\frac{x-\mu}{\sigma}\right)\right) \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\sigma^3\sqrt{2\pi}} (x-\mu) \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{\sigma^3\sqrt{2\pi}} (x-\mu) \times \left(-\frac{1}{2} \times 2 \times \frac{1}{\sigma}\left(\frac{x-\mu}{\sigma}\right)\right) \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} - \frac{1}{\sigma^3\sqrt{2\pi}} \times \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$$

$$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{1}{\sigma^5\sqrt{2\pi}} (x-\mu)^2 \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} - \frac{1}{\sigma^3\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$$

Points of Inflection exist when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$:

$\frac{1}{\sigma^5\sqrt{2\pi}} (x-\mu)^2 \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} - \frac{1}{\sigma^3\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} = 0$

$\Rightarrow \frac{1}{\sigma^5\sqrt{2\pi}} (x-\mu)^2 - \frac{1}{\sigma^3\sqrt{2\pi}} = 0$

$\Rightarrow \frac{1}{\sigma^5} (x-\mu)^2 - \frac{1}{\sigma^3} = 0$

$\Rightarrow (x-\mu)^2 - \sigma^2 = 0$

$\Rightarrow (x-\mu)^2 = \sigma^2 \Rightarrow x - \mu = \pm\sigma \Rightarrow x = \mu \pm \sigma$

46 a)

$\mathrm{P}(X < 70 \mid X < 80) = \frac{\mathrm{P}(X<70 \cap X<80)}{\mathrm{P}(X<80)} = \frac{\mathrm{P}(X<70)}{\mathrm{P}(X<80)}$

$= \frac{0.05692315}{0.9430769} = 0.0604$

46 b)

$\mathrm{P}(X > 70 \mid X < 76) = \frac{\mathrm{P}(X>70 \cap X<76)}{\mathrm{P}(X<76)} = \frac{\mathrm{P}(70<X<76)}{\mathrm{P}(X<76)}$

$= \frac{0.567162}{0.6240852} = 0.9088$

46 c)

$\mathrm{P}(X < 77 \mid X > 73) = \frac{\mathrm{P}(X<77 \cap X>73)}{\mathrm{P}(X>73)} = \frac{\mathrm{P}(73<X<77)}{\mathrm{P}(X>73)}$

$= \frac{0.4729107}{0.7364554} = 0.6421$

46 d)

$\mathrm{P}(X > 81 \mid X > 80) = \frac{\mathrm{P}(X>81 \cap X>80)}{\mathrm{P}(X>80)} = \frac{\mathrm{P}(X>81)}{\mathrm{P}(X>80)}$

$= \frac{0.02888979}{0.05692315} = 0.5075$

47)

$X \sim \mathrm{N}(75, 5^2)$

$\mathrm{P}(X > 80 \mid X > 76) = \frac{\mathrm{P}(X>80 \cap X>76)}{\mathrm{P}(X>76)} = \frac{\mathrm{P}(X>80)}{\mathrm{P}(X>76)}$

$= \frac{0.1586553}{0.4207403} = 0.3771$

48)

$X \sim \mathrm{N}(33, 3.5^2)$

$\mathrm{P}(X < 36 \mid X > 34) = \frac{\mathrm{P}(X<36 \cap X>34)}{\mathrm{P}(X>34)} = \frac{\mathrm{P}(34<X<36)}{\mathrm{P}(X>34)}$

$= \frac{0.1918655}{0.3875485} = 0.4951$

49 a)

$X \sim \mathrm{N}(49, 9)$

$\mathrm{P}(X < 47) = 0.2525$

49 b)

$Y \sim \mathrm{B}(12, 0.2525)$

$\mathrm{P}(Y \geq 4) = 1 - \mathrm{P}(Y \leq 3) = 0.3590$

50 a i)

$X \sim \mathrm{N}(10, 1.2^2)$

$\mathrm{P}(X = 11) = 0$

50 a ii)

$\mathrm{P}(X > 11) = 0.2023$

50 b)

$Y \sim \mathrm{B}(8, 0.2023)$

$\mathrm{P}(Y \geq 2) = 1 - \mathrm{P}(Y \leq 1) = 0.5034$

51)

$W \sim \mathrm{N}(5.7, 0.9^2)$

$\mathrm{P}(W < 4.8) = 0.1586553$

$\mathrm{P}(4.8 \leq 4.8 \leq 6.2) = 0.5520874$

$\mathrm{P}(W > 6.2) = 0.2892574$

$$150 \times 0.1586553 \times £0.11 + 150 \times 0.5520874 \times £0.25 + 150 \times 0.2892574 \times £0.21 = £32.43$$

52 a)

Total Area $= 10 \times 2 + 5 \times 4 + 5 \times 6 + 5 \times 9 + 10 \times 5 + 15 \times 1 = 180$

$\mathrm{P}(x < 170) = \frac{10\times2+5\times4+5\times6}{180} = \frac{70}{180} = \frac{7}{18}$

$= 0.3889$ to 4dp

52 b)

$\bar{x} = 172.3611$

$s = 10.05244$

52 c)

$Y \sim \mathrm{N}(172.3611, 10.05244^2)$

$\mathrm{P}(Y < 170) = 0.4072$

The data in the histogram appears fairly symmetric, and the percentage difference in the probabilities is $\frac{0.4072-0.3889}{0.3889} \times 100 = 4.71\%$. So modelling the data as a normal distribution is not very accurate but might be suitable for rough calculations.

53 a)

Total Area $= 5 \times 3 + 5 \times 5 + 10 \times 8 + 5 \times 7 + 15 \times 4 + 10 \times 1 = 225$

$\mathrm{P}(x < 170) = \frac{5\times3+5\times5+10\times8}{225} = \frac{120}{225} = \frac{8}{15}$

$= 0.5333$ to 4dp

53 b)

$\bar{x} = 170.5$

$s = 10.83305$

53 c)

$Y \sim \mathrm{N}(170.5, 10.83305^2)$

$\mathrm{P}(Y < 170) = 0.4816$

The data in the histogram appears positively skewed and the percentage difference in the probabilities is $\frac{0.5333-0.4816}{0.5333} \times 100 = 9.69\%$. So it appears that modelling the data as a normal distribution would not be appropriate.

54)

For a binomial distribution $X \sim \mathrm{B}(n, p)$ to be approximated by a normal distribution $Y \sim \mathrm{N}(\mu, \sigma^2)$, n must be large and p must be close to 0.5. It can also be shown that a normal distribution can approximate a binomial distribution if both $np > 5$ and $n(1 - p) > 5$

55 a)

$\mu = 500 \times 0.48 = 240$

$\sigma^2 = 500 \times 0.48 \times 0.52 = 124.8$

$Y \sim \mathrm{N}(240, 124.8)$

500 is large and 0.48 is close to 0.5

Check: $500 \times 0.48 = 240 > 5$ and $500 \times 0.52 = 260 > 5$

So a normal approximation would be appropriate.

55 b)

$\mu = 4 \times 0.5 = 2$

$\sigma^2 = 4 \times 0.5 \times 0.5 = 1$

$Y \sim \mathrm{N}(2, 1)$

2 is small but 0.5 is exactly what is required

Check: $2 \times 0.5 = 1 < 5$ and $2 \times 0.5 < 5$

So a normal approximation would not be appropriate.

55 c)

$\mu = 150000 \times 0.001 = 150$

$\sigma^2 = 150000 \times 0.001 \times 0.999 = 149.85$

$Y \sim \mathrm{N}(150, 149.85)$

150000 is large but 0.001 is not close to 0.5

Check: $150000 \times 0.001 = 150 > 5$ and $150000 \times 0.999 = 149850 > 5$

So a normal approximation might be appropriate, depending on the context, as a relatively large proportion (11%) would give negative values.

56 a)

$\mathrm{P}(X = 2) = \frac{160}{360} = \frac{4}{9}$

$X \sim \mathrm{B}(2000, \frac{4}{9})$

2000 is large and $\frac{4}{9} = 0.4444$ to 4dp is relatively close to 0.5

Check: $2000 \times \frac{4}{9} = \frac{8000}{9} > 5$

and $2000 \times \frac{5}{9} = \frac{10000}{9} > 5$

So a normal approximation would be appropriate.

56 b)

$\mu = 2000 \times \frac{4}{9} = \frac{8000}{9} = 889$ to 3sf

$\sigma^2 = 2000 \times \frac{4}{9} \times \frac{5}{9} = \frac{40000}{81} = 494$ to 3sf

$Y \sim \mathrm{N}(889, 494)$

$\mathrm{P}(Y > a) = 0.2 \Rightarrow a = 908$

56 c)

$X \sim \mathrm{B}(2000, \frac{4}{9})$

$\mathrm{P}(X > 908) = 0.1887$

57)

$\mathrm{P}(X > a) = b \Rightarrow Z = \frac{a-300}{20}$

$\mathrm{P}(Y < a) = b \Rightarrow Z = \frac{a-400}{30}$

$\frac{a-300}{20} = -\frac{a-400}{30} \Rightarrow 30a - 9000 = -20a + 8000$

$\Rightarrow 50a = 17000 \Rightarrow a = 340$

$\therefore b = \mathrm{P}(X > 340) = 0.0228$ to 4dp

58)

$\mathrm{P}(X > a) = b \Rightarrow Z = \frac{a-16.5}{2.3}$

$\mathrm{P}(Y < a) = b \Rightarrow Z = \frac{a-20.2}{2.6}$

$\frac{a-16.5}{2.3} = -\frac{a-20.2}{2.6} \Rightarrow 2.6a - 42.9 = -2.3a + 46.46$

$\Rightarrow 4.9a = 89.36 \Rightarrow a = \frac{4468}{245} = 18.2$ to 3sf

$\therefore b = \mathrm{P}(X > \frac{4468}{245}) = 0.2251$ to 4dp

59)

$\mathrm{P}(X < a) = b \Rightarrow Z = \frac{a-50}{2}$

$\mathrm{P}(Y > a) = b \Rightarrow Z = \frac{a-60}{3}$

$\frac{a-50}{2} = -\frac{a-60}{3} \Rightarrow 3a - 150 = -2a + 120$

$\Rightarrow 5a = 270 \Rightarrow a = 54$

$\therefore b = \mathrm{P}(X < 54) = 0.9772$ to 4dp

60)

$\mathrm{P}(X < a - 15) = b \Rightarrow Z = \frac{(a-15)-200}{5}$

$\mathrm{P}(Y > a + 15) = b \Rightarrow Z = \frac{(a+15)-250}{7}$

$\frac{a-215}{5} = -\frac{a-235}{7} \Rightarrow 7a - 1505 = -5a + 1175$

$\Rightarrow 12a = 2680 \Rightarrow a = \frac{670}{3} = 223$ to 3sf

$\therefore b = \mathrm{P}(X < \frac{670}{3} - 15) = 0.9522$ to 4dp

61 a)

$\mathrm{P}(X \leq 300) \approx \mathrm{P}(Y < 300.5) = 0.1737$

61 b)

$\mathrm{P}(X > 320) \approx \mathrm{P}(Y > 320.5) = 0.2437$

61 c)

$\mathrm{P}(X < 311) \approx \mathrm{P}(Y < 310.5) = 0.4512$

61 d)

$\mathrm{P}(X \geq 295) \approx \mathrm{P}(Y > 294.5) = 0.9236$

61 e)

$\mathrm{P}(X = 312) \approx \mathrm{P}(311.5 < Y < 312.5) = 0.0326$

62 a)

$\mathrm{P}(X \leq 200) = 0.1500297$

62 b)

$Y \sim \mathrm{N}(210, 84)$

$\mathrm{P}(Y < 200.5) = 0.1499763$

62 c)

$\%$ error $= \frac{|0.1500297 - 0.1499763|}{0.1500297} \times 100 = 0.0356\%$

63 a)

$\mathrm{P}(X < 260) = \mathrm{P}(X \leq 259) = 0.6886616$

63 b)

$Y \sim \mathrm{N}(253.8, 134.514)$

$\mathrm{P}(Y < 259.5) = 0.6884506$

63 c)

$\%$ error $= \frac{|0.6886616 - 0.6884506|}{0.6886616} \times 100 = 0.0306\%$

64 a)

$\mathrm{P}(X \geq 1440) = 1 - \mathrm{P}(X \leq 1439) = 0.02639502$

64 b)

$Y \sim \mathrm{N}(1400, 420)$

$\mathrm{P}(Y > 1439.5) = 0.02696474$

64 c)

$\%$ error $= \frac{|0.02639502 - 0.02696474|}{0.02639502} \times 100 = 2.16\%$ (3 SF)

65 a i)

$X \sim \mathrm{B}(5000, 0.045)$

$\mathrm{P}(X > 230) = 1 - \mathrm{P}(X \leq 230) = 0.3505$

65 a ii)

$\mathrm{P}(X > 230 \mid X < 250) = \frac{\mathrm{P}(X > 230 \cap X < 250)}{\mathrm{P}(X < 250)}$

$= \frac{\mathrm{P}(230 < X < 250)}{\mathrm{P}(X < 250)} = \frac{0.3013867}{0.9509236} = 0.3169$

65 b)

$Y \sim \mathrm{N}(225, 214.875)$

$\mathrm{P}(X > 230) \approx \mathrm{P}(Y > 230.5) = 0.3538$

$\mathrm{P}(X > 230 \mid X < 250) = \frac{\mathrm{P}(230<X<250)}{\mathrm{P}(X<250)}$

$\approx \frac{\mathrm{P}(230.5<Y<249.5)}{\mathrm{P}(X<249.5)} = \frac{0.3064295}{0.9526759} = 0.3217$